OCCUPATION SUMS OF THE HARMONIC DESCENT CHAIN AND SLOWLY DECAYING WEIGHTS

YITZCHAK SHMALO

ABSTRACT. The harmonic descent chain jumps from a state $m \geq 2$ to $m - d$ with probability proportional to $1/d$. Started at n, it records the clade sizes on the path from the root to a uniform leaf of the critical beta-splitting tree with n leaves. Let $a(n, i)$ be the probability that the chain started at n visits i, and let $a(i)$ be its limit as $n \to \infty$. Aldous and Janson asked whether the occupation sums $\sum_{i \leq n} a(n, i) f(i)$ converge to $\sum_i a(i) f(i)$ for every non-negative weight f of polynomial growth for which the limit sum is finite. Brandenberger, Chin and Mossel proved the convergence when $f(i) = O(i^{-\varepsilon})$. We prove it whenever $f(i) = O(1/\log i)$, and we show that this rate is the threshold: for every φ tending to infinity there is a weight $f(i) \leq \varphi(i)/\log i$, tending to zero, with a finite limit sum and unbounded occupation sums. The companion statement for infinite limit sums fails for weights of linear growth, where the ratio of the two sides tends to $\pi^2/6$, and for a bounded weight.

1. INTRODUCTION

The harmonic descent chain is the Markov chain $(X_t)_{t \geq 0}$ on $\{1, 2, \dots\}$ which, from a state $m \geq 2$, jumps to $m - d$ with probability

$$p(m, m - d) = \frac{1}{d\, h_{m-1}} \qquad (1 \leq d \leq m - 1), \tag{1.1}$$

and which stays at 1 once it is there. Here $h_r = \sum_{j=1}^{r} 1/j$ is the rth harmonic number, and $h_0 = 0$. We write $\mathbb{P}_n$ and $\mathbb{E}_n$ for probability and expectation when $X_0 = n$. Asymptotic statements refer to the limit of the variable named, $n \to \infty$ unless another is stated: $x_n \sim y_n$ means $x_n/y_n \to 1$, $x_n = o(y_n)$ means $x_n/y_n \to 0$, and $x_n = O(y_n)$ means $|x_n| \leq C y_n$ for a constant C that does not depend on n. Started at n, the chain records the clade sizes on the path from the root to a uniformly chosen leaf of the discrete-time critical beta-splitting tree with n leaves [3, 1, 2].

The occupation probability $a(n, i)$ is the probability that the chain started at n visits the state i, so that $a(n, n) = a(n, 1) = 1$. For every fixed $i \geq 2$ the limit

$$a(i) = \lim_{n \to \infty} a(n, i) = \frac{6\, h_{i-1}}{\pi^2 (i - 1)} \tag{1.2}$$

exists [1, Theorem 18], [2]. Iksanov [7] gave a second proof through regenerative composition structures, and Chorbadzhiyska, Minchev and Savov [5] express the occupation probabilities, for the whole beta-splitting family, through the potential measure of a subordinator. Aldous and Janson [1, Section 4.7, Open Problem 8] asked for explicit bounds on $|a(n, i) - a(i)|$, and in particular for a proof of the following statement.

Ansatz 20 of [1]**.** For a non-negative sequence $(f(j),\, j \geq 2)$ such that $f(j) = O(j^k)$ for some $k < \infty$:

Date: 12 November 2026.

2020 *Mathematics Subject Classification.* Primary 60J10; Secondary 60J80, 60C05, 05C05.

Key words and phrases. harmonic descent chain, critical beta-splitting tree, occupation probability, harmonic number.

(i) if $\sum_{i\ge2} a(i)f(i) < \infty$ then $\sum_{i=2}^{n} a(n,i)f(i) \to \sum_{i\ge2} a(i)f(i)$;
(ii) if $\sum_{i\ge2} a(i)f(i) = \infty$ then $\sum_{i=2}^{n} a(n,i)f(i) \sim \sum_{i=2}^{n} a(i)f(i)$.

We call a non-negative sequence f on $\{2,3,\dots\}$ a weight, $\sum_{i=2}^{n} a(n,i)f(i)$ its occupation sum at n, and $\sum_{i\ge2} a(i)f(i)$, finite or not, its limit sum. The Ansatz is the step that turns the limit (1.2) into asymptotics for additive statistics of the tree, such as the balance indices of [1, Section 4.9], where it is noted that the method can work only for indices that are not sensitive to the structure near the root.

Brandenberger, Chin and Mossel [4] proved part (i) for every weight with $f(i) = O(i^{-\varepsilon})$ for some $\varepsilon > 0$. Their main theorem contains the monotonicity conjectured in [3]: for each $i \ge 2$ the sequence $a(n,i)$, $n \ge i$, is strictly decreasing [4, Theorem 1.1(i)]. It also gives the rate $a(n,i) - a(i) = n^{-\gamma_*+o(1)}$ for fixed i, with $\gamma_* = 1.567\dots$, and two-sided bounds of this kind for $i = o(n/\log^3 n)$ [4, (1.6), (1.7)]. Svihla [8] gives an exact asymptotic expansion of $a(n,i)$ for $i = o(\sqrt{n})$. These results concern states i far below the starting point n. The Ansatz also involves the states i of the same order as n, and these are the subject of this paper.

Part (i) restricts the decay of the weight. The occupation sum at n contains the term $a(n,n)f(n) = f(n)$, and by Fatou's lemma and (1.2) the other terms satisfy

$$\liminf_{n\to\infty} \sum_{i=2}^{n-1} a(n,i)f(i) \ge \sum_{i\ge2} a(i)f(i).$$

Part (i) can therefore hold only for weights with $f(n) \to 0$, and the question is how slowly f may decay. The answer is the rate $1/\log i$.

Theorem 1.1. *Let f be a weight with $f(i) \le K/\log i$ for all $i \ge 2$ and some constant $K < \infty$, and suppose that $\sum_{i\ge2} a(i)f(i) < \infty$. Then*

$$\sum_{i=2}^{n} a(n,i)f(i) \to \sum_{i\ge2} a(i)f(i) \qquad (n \to \infty).$$

Theorem 1.2. *Let φ be a positive sequence on $\{2,3,\dots\}$ with $\varphi(i) \to \infty$. There are a weight f and integers $N_1 < N_2 < \cdots$ such that $f(i) \le \varphi(i)/\log i$ for all $i \ge 2$, $f(i) \to 0$, $\sum_{i\ge2} a(i)f(i) < \infty$, $f(N_j) = 0$ for every j, and*

$$\sum_{i=2}^{N_j} a(N_j,i)f(i) \to \infty \qquad (j \to \infty).$$

The weight of Theorem 1.2 is bounded, so it satisfies the growth condition of the Ansatz with $k = 0$, and part (i) fails for it. Since $f(N_j) = 0$, the same holds for the sums over $2 \le i \le n-1$.

Part (ii) fails for weights that grow, and also for a bounded weight. Let

$$g(i) = \frac{i-1}{h_{i-1}} \qquad (i \ge 2), \tag{1.3}$$

which by (1.1) is the expected size of the jump out of the state i.

Proposition 1.3. (a) *For every $n \ge 2$,*

$$\sum_{i=2}^{n} a(n,i)g(i) = n-1 \qquad \text{and} \qquad \sum_{i=2}^{n} a(i)g(i) = \frac{6}{\pi^2}(n-1).$$

If f is a weight with $f(i)/i \to c$ for some $c \in (0,\infty)$, then

$$\sum_{i=2}^{n} a(n,i)f(i) \sim c\,n\log n \qquad \text{and} \qquad \sum_{i=2}^{n} a(i)f(i) \sim \frac{6}{\pi^2}\,c\,n\log n.$$

(b) *There is a weight f with values in $\{0,1\}$ such that $\sum_{i\ge 2} a(i)f(i) = \infty$ and*

$$\limsup_{n\to\infty} \frac{\sum_{i=2}^{n} a(n,i)f(i)}{\sum_{i=2}^{n} a(i)f(i)} = \infty.$$

In part (a) the two sides of part (ii) of the Ansatz differ by the factor $\pi^2/6$. A weight of this kind, with $f(j) \sim j/4$, is the one to which the Ansatz is applied in [1, Section 4.9.2].

Remark 1.4. Part (i) holds for every non-increasing weight with a finite limit sum. Indeed, (1.2) gives $a(k) \ge 6/(\pi^2(k-1))$, so for a non-increasing weight

$$f(i)\,\frac{6}{\pi^2}\,\log i \le f(i)\sum_{k=2}^{i} a(k) \le \sum_{k\ge 2} a(k)f(k),$$

and Theorem 1.1 applies.

All of these statements come from counting the visits of the chain to the states just below its starting point. For integers $1 \le L \le n-2$ we call

$$W(n,L) = \{n-L,\dots,n-1\}$$

the window of width L below n, and we write

$$V(n,L) = \sum_{i\in W(n,L)} a(n,i) \tag{1.4}$$

for the expected number of visits of the chain started at n to this window. Lemma 2.1 and Corollary 3.2 give

$$\frac{\log n}{10\log(8n/L)} \le V(n,L) \le \frac{h_L}{h_{n-L-1}-h_L}$$

when $8 \le L < n/2$ and $h_{n-1} \ge 2\log(8n/L)$. Fix $j \ge 2$ and let $L = \lfloor n/2^j \rfloor$. For large n both conditions hold, and the window $W(n,L)$ receives of order $(\log n)/j$ visits, while by (1.2) the limits $a(i)$ of its states add up to $(6/\pi^2)(-\log(1-2^{-j}) + o(1))\log n$, which is of order $2^{-j}\log n$. A weight of size $K/\log n$ on such a window adds a bounded amount to the occupation sum, while a weight of size $\varphi(n)/\log n$ with $\varphi(n) \to \infty$ can add an unbounded one.

Section 2 contains the upper bounds and the proof of Theorem 1.1, Section 3 the lower bound, and Section 4 the proofs of Theorem 1.2 and Proposition 1.3. Section 5 reports a numerical check of the bounds, and the appendix lists the notation.

2. Upper bounds and the proof of Theorem 1.1

The chain decreases strictly until it reaches 1, so it visits each state $i \ge 2$ at most once. Hence, for a set B of states in $\{2,\dots,n-1\}$,

$$\sum_{i\in B} a(n,i) = \mathbb{E}_n\big[\text{number of } t \ge 1 \text{ with } X_t \in B\big]. \tag{2.1}$$

We use the elementary bounds

$$\log(r+1) \le h_r \le 1+\log r \quad (r\ge 1) \qquad \text{and} \qquad \frac{s-r}{s} \le h_s - h_r \le \log\frac{s}{r} \quad (1\le r\le s). \tag{2.2}$$

Lemma 2.1. *Let $L \ge 1$ and $n \ge 2L+2$. Then*

$$V(n,L) \le \frac{h_L}{h_{n-L-1}-h_L}.$$

Proof. Let m be a state with $n-L \le m \le n$. The next state $m-d$ lies in $W(n,L)$ exactly when $d \le m-(n-L)$, and by (1.1) this has probability

$$\frac{h_{m-n+L}}{h_{m-1}} \le \frac{h_L}{h_{n-L-1}} =: \rho,$$

because $m-n+L \le L$ and $m-1 \ge n-L-1$. Since $n-L-1 > L$ we have $\rho < 1$. The chain is decreasing, so after it has left $W(n,L) \cup \{n\}$ it does not return. The number of its visits to $W(n,L)$ is therefore at least k exactly when $X_1, \dots, X_k$ lie in $W(n,L)$, which by the Markov property has probability at most ρ^k. By (2.1),

$$V(n,L) \le \sum_{k \ge 1} \rho^k = \frac{\rho}{1-\rho} = \frac{h_L}{h_{n-L-1} - h_L}. \qquad \square$$

The next lemma bounds the occupation probabilities of a fixed state i summed over the starting points just above i.

Lemma 2.2. *Let $i \ge 2$ and $0 \le t < i$. Then*

$$S_t(i) := \sum_{\ell=i}^{i+t} a(\ell, i) \le \frac{h_i}{h_i - h_t}.$$

Proof. Conditioning on the first jump gives, for $s \ge 1$,

$$a(i+s,i) = \frac{1}{h_{i+s-1}} \sum_{d=1}^{s} \frac{a(i+s-d,i)}{d} \le \frac{1}{h_i} \sum_{d=1}^{s} \frac{a(i+s-d,i)}{d},$$

since $h_{i+s-1} \ge h_i$. Summing over $1 \le s \le t$ and collecting the terms with $s-d=r$,

$$S_t(i) - 1 \le \frac{1}{h_i} \sum_{r=0}^{t-1} a(i+r,i) \sum_{d=1}^{t-r} \frac{1}{d} = \frac{1}{h_i} \sum_{r=0}^{t-1} a(i+r,i)\, h_{t-r} \le \frac{h_t}{h_i}\, S_t(i).$$

As $t < i$ we have $h_t < h_i$, and the bound follows. $\square$

Lemma 2.3. *Let $i \ge 2$, $0 \le t < i$ and $n \ge i+t$. Then*

$$a(n,i) \le \frac{h_i}{(t+1)(h_i - h_t)}, \qquad \text{and if } t \ge 1 \text{ also} \qquad a(n,i) \le \frac{h_t}{t\,(h_i - h_t)}.$$

Proof. By [4, Theorem 1.1(i)] we have $a(n,i) \le a(m,i)$ for $i \le m \le n$. Hence $(t+1)\,a(n,i) \le S_t(i)$ and $t\,a(n,i) \le S_t(i) - a(i,i) = S_t(i) - 1$, and Lemma 2.2 gives both bounds. $\square$

Corollary 2.4. *Let $i \ge 2$.*

(a) *If $n \ge i + \lfloor i/2 \rfloor$, then $a(n,i) \le 4h_i/i \le \pi^2 a(i)$.*

(b) *If $0 < \varepsilon \le 1/2$ and $n - i \ge \varepsilon n$, then $a(n,i) \le \dfrac{\pi^2}{2\varepsilon}\, a(i)$.*

Proof. For $0 \le t \le i/2$ the bound (2.2) gives $h_i - h_t \ge (i-t)/i \ge 1/2$, so the first bound of Lemma 2.3 reads

$$a(n,i) \le \frac{2h_i}{t+1} \qquad (n \ge i+t,\ 0 \le t \le i/2). \tag{2.3}$$

For (a) take $t = \lfloor i/2 \rfloor$, so that $t+1 > i/2$ and $a(n,i) \le 4h_i/i$. Since $h_{i-1} \ge 1$ we have $h_i \le h_{i-1}(1+1/i)$, and therefore

$$\frac{4h_i}{i} \le \frac{4(i+1)}{i^2}\, h_{i-1} \le \frac{6\, h_{i-1}}{i-1} = \pi^2 a(i), \tag{2.4}$$

the second inequality being $4(i^2-1) \le 6i^2$. For (b), if $n \ge i + \lfloor i/2 \rfloor$ the claim follows from (a) because $2\varepsilon \le 1$. Otherwise $t = n-i < i/2$, and (2.3) with $t+1 > \varepsilon n \ge \varepsilon i$ gives $a(n,i) \le 2h_i/(\varepsilon i)$, which is at most $\pi^2 a(i)/(2\varepsilon)$ by (2.4). □

Proof of Theorem 1.1. Fatou's lemma and (1.2) give $\liminf_n \sum_{i=2}^n a(n,i)f(i) \ge \sum_{i\ge 2} a(i)f(i)$, so we have to bound the upper limit. Fix $\varepsilon \in (0, 1/3]$ and an integer $M \ge 2$, and let $L = \lfloor \varepsilon n \rfloor$. For $n \ge \max(6, M/\varepsilon)$ we have $n \ge 2L+2$ and $M \le n-L-1$, and we split the occupation sum at n into the ranges $2 \le i < M$ and $M \le i \le n-L-1$, the window $W(n,L)$, and the term $i = n$.

By (1.2) the sum over $2 \le i < M$ tends to $\sum_{i=2}^{M-1} a(i)f(i)$. For $M \le i \le n-L-1$ we have $n - i \ge L+1 > \varepsilon n$, so Corollary 2.4(b) gives

$$\sum_{i=M}^{n-L-1} a(n,i)f(i) \le \frac{\pi^2}{2\varepsilon} \sum_{i\ge M} a(i)f(i).$$

On the window, $f(i) \le K/\log(n-L)$, so by Lemma 2.1

$$\sum_{i \in W(n,L)} a(n,i)f(i) \le \frac{K}{\log(n-L)} \cdot \frac{h_L}{h_{n-L-1} - h_L}.$$

As $n \to \infty$ we have $h_L/\log(n-L) \to 1$ and $h_{n-L-1} - h_L \to \log((1-\varepsilon)/\varepsilon)$, which is at least $\log 2$. The last term is $f(n) \le K/\log n$. Altogether,

$$\limsup_{n\to\infty} \sum_{i=2}^{n} a(n,i)f(i) \le \sum_{i=2}^{M-1} a(i)f(i) + \frac{\pi^2}{2\varepsilon} \sum_{i\ge M} a(i)f(i) + \frac{K}{\log((1-\varepsilon)/\varepsilon)}.$$

Letting $M \to \infty$ and then $\varepsilon \to 0$ gives $\limsup_n \sum_{i=2}^n a(n,i)f(i) \le \sum_{i\ge 2} a(i)f(i)$. □

Remark 2.5. Let f be a weight with $f(i) \le K/\log i$ and $\sum_{i\ge 2} a(i)f(i) = \infty$. Then

$$1 \le \liminf_{n\to\infty} \frac{\sum_{i=2}^n a(n,i)f(i)}{\sum_{i=2}^n a(i)f(i)} \le \limsup_{n\to\infty} \frac{\sum_{i=2}^n a(n,i)f(i)}{\sum_{i=2}^n a(i)f(i)} \le \frac{3\pi^2}{2}.$$

The lower bound holds because $a(n,i) \ge a(i)$ for $n \ge i$, by [4, Theorem 1.1(i)]. For the upper bound take $\varepsilon = 1/3$ and $M = 2$ in the proof above: the terms with $i \le n-L-1$ are at most $(3\pi^2/2)\, a(i)f(i)$, and the window and the term $i = n$ contribute a bounded amount. We do not know whether the ratio tends to 1 for every such weight.

Remark 2.6. For $1 \le t \le i^\theta$ with a fixed $\theta < 1$, the second bound of Lemma 2.3 is of order $h_t/(t\,h_i)$. In a range of this kind, and for large i, [4, Theorem 1.1(iii)] gives the bound $a(i+t,i) \le C/(t\,h_i)$, which is smaller by a factor of order h_t. Lemma 2.3 has explicit constants and covers every $t < i$; the proof of Theorem 1.1 uses it with t of order i.

3. A lower bound for the visits to a window

For $s \ge 1$ let $D_s = X_{s-1} - X_s$ be the sth jump of the chain, so that $D_s = 0$ when $X_{s-1} = 1$.

Lemma 3.1. *Let $n \ge 4$, let y, L, T be integers with $1 \le y \le L \le n-2$ and $T \ge 1$, and put $q = h_y/h_{n-1}$. Then*

$$V(n,L) \ge T\Big(q^T - \frac{T\,y}{L\,h_{n-L-1}}\Big).$$

Proof. Let $2 \le m \le n$. Given $X_{s-1} = m$, the probability that $D_s \le y$ is $h_y/h_{m-1} \ge q$ if $y \le m-1$, and is 1 otherwise; given $X_{s-1} = 1$ it is 1. By the Markov property the event $E = \{D_1 \le y, \dots, D_T \le y\}$ has

$$\mathbb{P}_n(E) \ge q^T.$$

Let $Z = \sum_{s=1}^{T} D_s\,\mathbf{1}\{D_s \le y,\ X_{s-1} \ge n-L\}$. Given $X_{s-1} = m \ge n-L$, the expectation of $D_s\mathbf{1}\{D_s \le y\}$ is $\min(y, m-1)/h_{m-1} \le y/h_{n-L-1}$ by (1.1), so

$$\mathbb{E}_n[Z] \le \frac{T\,y}{h_{n-L-1}}.$$

On the event $E \cap \{X_T < n-L\}$ let $\tau \le T$ be the first time with $X_\tau < n-L$. For $s \le \tau$ we have $X_{s-1} \ge n-L$ and $D_s \le y$, so $Z \ge D_1 + \cdots + D_\tau = n - X_\tau > L$. By Markov's inequality,

$$\mathbb{P}_n(E \cap \{X_T < n-L\}) \le \frac{\mathbb{E}_n[Z]}{L} \le \frac{T\,y}{L\,h_{n-L-1}}.$$

On the event $E \cap \{X_T \ge n-L\}$ the chain has not reached 1 by time T, because $n-L \ge 2$. Hence $n-1 \ge X_1 > \cdots > X_T \ge n-L$, and the chain makes at least T visits to $W(n,L)$. By (2.1),

$$V(n,L) \ge T\,\mathbb{P}_n(E \cap \{X_T \ge n-L\}) \ge T\Big(q^T - \frac{T\,y}{L\,h_{n-L-1}}\Big). \qquad \square$$

Corollary 3.2. *Let $8 \le L \le n/2$, put $\lambda = \log(8n/L)$, and suppose that $h_{n-1} \ge 2\lambda$. Then*

$$V(n,L) \ge \frac{h_{n-1}}{10\,\lambda} \ge \frac{\log n}{10\log(8n/L)}.$$

Proof. Write $H = h_{n-1}$, and apply Lemma 3.1 with $T = \lfloor H/\lambda \rfloor$ and $y = \lceil L/8 \rceil$. Since $H/\lambda \ge 2$ and $\lfloor x \rfloor \ge 2x/3$ for $x \ge 2$, we have

$$\frac{2H}{3\lambda} \le T \le \frac{H}{\lambda}.$$

By (2.2) and $y \ge L/8$ we have $H - h_y \le \log((n-1)/y) \le \lambda$, so $q \ge 1-x$ with $x = \lambda/H \le 1/2$. The function $x \mapsto (1-x)^{1/x}$ is decreasing on $(0,1)$ and equals $1/4$ at $x = 1/2$, and $T \le 1/x$, so

$$q^T \ge (1-x)^{1/x} \ge \frac{1}{4}.$$

Next, $y/L \le 2/9$: for $L = 8$ we have $y = 1$, and for $L \ge 9$ we have $y \le (L+7)/8 \le 2L/9$. Since $L \le n/2$, we have $\lambda \ge \log 16$ and $H \ge 2\log 16 > 5.5$, and by (2.2)

$$h_{n-L-1} = H - \sum_{j=n-L}^{n-1} \frac{1}{j} \ge H - \frac{L}{n-L} \ge H - 1, \qquad \text{so} \qquad \frac{H}{h_{n-L-1}} \le \frac{5.5}{4.5} < 1.23.$$

Therefore

$$\frac{T\,y}{L\,h_{n-L-1}} \le \frac{2}{9\lambda} \cdot \frac{H}{h_{n-L-1}} \le \frac{2 \times 1.23}{9\log 16} < 0.099,$$

and Lemma 3.1 gives

$$V(n,L) \ge \frac{2H}{3\lambda}\,(0.25 - 0.099) > \frac{H}{10\,\lambda}.$$

The second inequality of the corollary is $h_{n-1} \ge \log n$. $\square$

4. Proofs of Theorem 1.2 and Proposition 1.3

By (1.2), (2.2) and $i - 1 \ge i/2$,

$$\frac{6}{\pi^2} \cdot \frac{\log i}{i} \le a(i) \le \frac{12}{\pi^2}\Big(1 + \frac{1}{\log 2}\Big)\frac{\log i}{i} \le \frac{3\log i}{i} \qquad (i \ge 2). \tag{4.1}$$

Proof of Theorem 1.2. Let $\psi(i) = \inf_{k \ge i} \varphi(k)$. Then $\psi \le \varphi$, and ψ is positive, non-decreasing and tends to infinity. Choose integers $N_1 < N_2 < \cdots$ such that for every $j \ge 1$

$$\psi(\lceil N_j/2 \rceil) \ge j^2, \qquad \log N_j \ge \max\bigl(2(j+4),\, j^3\bigr), \qquad N_{j+1} \ge 2N_j + 2.$$

Let $L_j = \lfloor N_j/2^j \rfloor$ and $B_j = W(N_j, L_j)$, and define

$$f(i) = \frac{j^2}{\log N_j} \quad (i \in B_j,\ j \ge 1), \qquad f(i) = 0 \quad \text{for all other } i.$$

Every $i \in B_j$ satisfies $N_j/2 \le i \le N_j - 1$, and $N_j/2 > N_{j-1}$ for $j \ge 2$. The windows B_j are therefore disjoint, none of them contains any N_k, and $f(N_k) = 0$ for every k.

Decay. For $i \in B_j$ we have $i \ge \lceil N_j/2 \rceil$, so $\varphi(i) \ge \psi(i) \ge j^2$, and $\log i \le \log N_j$. Hence $f(i) \le \varphi(i)/\log i$. Moreover $f(i) \le j^2/j^3 = 1/j$ on B_j, so $f(i) \to 0$.

Finite limit sum. By (4.1), each $i \in B_j$ has $a(i) \le 3 \log N_j/(N_j/2)$, and B_j has $L_j \le N_j/2^j$ elements. Hence

$$\sum_{i \in B_j} a(i) f(i) \le \frac{N_j}{2^j} \cdot \frac{6 \log N_j}{N_j} \cdot \frac{j^2}{\log N_j} = \frac{6j^2}{2^j},$$

which is summable in j.

Unbounded occupation sums. We apply Corollary 3.2 with $n = N_j$ and $L = L_j$. Since $N_j \ge e^{2j+8} \ge 2^{j+3}$ we have $8 \le L_j \le N_j/2$ and $L_j \ge N_j/2^{j+1}$, so $\lambda_j := \log(8N_j/L_j) \le (j+4)\log 2$ and $h_{N_j - 1} \ge \log N_j \ge 2(j+4) \ge 2\lambda_j$. The corollary gives

$$\sum_{i=2}^{N_j} a(N_j, i) f(i) \ge \frac{j^2}{\log N_j} V(N_j, L_j) \ge \frac{j^2}{\log N_j} \cdot \frac{\log N_j}{10\,(j+4)\log 2} = \frac{j^2}{10\,(j+4)\log 2},$$

which tends to infinity. □

Proof of Proposition 1.3(a). The chain started at n reaches 1 after at most $n-1$ jumps, and its jumps add up to $n-1$. Given $X_t = m \ge 2$, the expected size of the next jump is $\sum_{d=1}^{m-1} d\, p(m, m-d) = g(m)$, and each state $m \ge 2$ is visited at most once. Taking expectations,

$$n - 1 = \sum_{m=2}^{n} a(n, m)\, g(m).$$

In the same way, for $2 \le r \le n$ the sum $\sum_{m=2}^{r} a(n,m) g(m)$ is the expected total size of the jumps made from states $m \le r$, and these jumps add up to at most $r-1$. Hence

$$\sum_{m=2}^{r} a(n, m)\, g(m) \le r - 1 \qquad (2 \le r \le n). \tag{4.2}$$

The second identity of (a) holds because $a(i)g(i) = 6/\pi^2$ for every $i \ge 2$.

Now consider the weight $i - 1 = g(i) h_{i-1}$. On the one hand $\sum_{i=2}^{n} a(n,i)(i-1) \le h_{n-1}(n-1)$. On the other hand, by (4.2), for $2 \le r < n$

$$\sum_{i=2}^{n} a(n,i)(i-1) \ge h_r \sum_{i=r+1}^{n} a(n,i)\, g(i) \ge h_r\,(n - r).$$

With $r = \lceil n/\log n \rceil$ both bounds are $(1 + o(1))\, n \log n$. Since $g \ge 1$, we also have $\sum_{i=2}^{n} a(n,i) \le n - 1$, and therefore

$$\sum_{i=2}^{n} a(n,i)\, i \sim n \log n. \tag{4.3}$$

For the limits, (1.2) gives $a(i)\,i = (6/\pi^2)\,h_{i-1}\,(1 + 1/(i-1))$, and $\sum_{i=2}^{n} h_{i-1} = n\,h_{n-1} - (n-1)$, so

$$\sum_{i=2}^{n} a(i)\,i \sim \frac{6}{\pi^2}\,n \log n. \tag{4.4}$$

Finally let $f(i)/i \to c \in (0,\infty)$. Given $\eta > 0$ choose M with $|f(i) - ci| \le \eta\,i$ for $i \ge M$. Since $a(n,i) \le 1$,

$$\Big|\sum_{i=2}^{n} a(n,i) f(i) - c \sum_{i=2}^{n} a(n,i)\,i\Big| \le \eta \sum_{i=2}^{n} a(n,i)\,i + \sum_{i=2}^{M-1} |f(i) - ci|,$$

and (4.3) gives $\sum_{i=2}^{n} a(n,i) f(i) \sim c\,n \log n$. The same argument with $a(i) \le 1$ and (4.4) gives the second asymptotic relation. □

Proof of Proposition 1.3(b). For $j \ge 5$ let $N_j = 2^{j^2}$, $L_j = \lfloor N_j/j^2 \rfloor$ and $B_j = W(N_j, L_j)$, and let f be the indicator of $\bigcup_{j\ge5} B_j$. Every $i \in B_j$ satisfies $N_j/2 \le i < N_j$, and $N_j/2 > N_{j-1}$, so the windows B_j are disjoint. The window B_j has L_j elements and $N_j/(2j^2) \le L_j \le N_j/j^2$. By (4.1), and because $\log x/x$ is decreasing for $x \ge e$, each $i \in B_j$ has

$$\frac{6}{\pi^2} \cdot \frac{j^2 \log 2}{N_j} \le a(i) \le \frac{6\,j^2 \log 2}{N_j}, \qquad \text{so} \qquad \frac{3 \log 2}{\pi^2} \le \sum_{i \in B_j} a(i) \le 6 \log 2.$$

Hence $\sum_{i\ge2} a(i) f(i) = \infty$ and $\sum_{i=2}^{N_j} a(i) f(i) \le 6\,j \log 2$.

We apply Corollary 3.2 with $n = N_j$ and $L = L_j$. Here $8 \le L_j \le N_j/2$ and $\lambda_j := \log(8N_j/L_j) \le \log(16j^2)$, and $h_{N_j-1} \ge j^2 \log 2 \ge 2\log(16j^2)$ for $j \ge 5$. The corollary gives

$$\sum_{i=2}^{N_j} a(N_j, i) f(i) \ge V(N_j, L_j) \ge \frac{j^2 \log 2}{10 \log(16j^2)},$$

so the ratio in (b), taken at $n = N_j$, is at least $j/(60\log(16j^2))$, which tends to infinity. □

5. A numerical check

The program `code/check_bounds.py` in the repository of this paper evaluates the occupation probabilities in double-precision arithmetic with NumPy [6], from the recursion $a(n,i) = \sum_{m=i+1}^{n} a(n,m)/((m-i)\,h_{m-1})$ for $i < n$, which follows from the Markov property at the last state above i. For i in the window $W(n,L)$ only the states of the window and n itself enter the recursion, so $V(n,L)$ can be evaluated with $O(L^2)$ operations for very large n.

For $n = 4000$ the program finds $\sum_{i=2}^{n} a(n,i) g(i) = 3999.0000000001$, and the ratio of this sum to $\sum_{i=2}^{n} a(i) g(i)$ agrees with $\pi^2/6$ to twelve digits. For the weight $f(i) = i$ the occupation sum is 32930.33 against $n \log n = 33176.20$, and $\sum_{i=2}^{n} a(i)\,i = 19165.35$ against $(6/\pi^2)\,n \log n = 20168.71$. The largest value of $a(n,i)/a(i)$ over the states with $n \ge i + \lfloor i/2 \rfloor$ is 1.2211, below the constant π^2 of Corollary 2.4(a). The bounds of Lemmas 2.2 and 2.3 hold in every case computed, with equality in Lemma 2.2 at $t = 0$.

Table 1 compares $V(n,L)$ with the bounds of Lemma 2.1 and Corollary 3.2.

Acknowledgements and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science

n	L	$V(n,L)$	lower bound	upper bound
2^9	56	1.6200	–	2.2095
2^{10}	64	1.4066	–	1.7572
2^{14}	1024	2.2855	0.2119	2.7734
2^{16}	4096	2.7284	0.2405	3.2848
2^{18}	16384	3.1717	0.2690	3.7966
2^{22}	16384	1.7413	0.2076	1.8554
2^{30}	16384	0.9047	–	0.9271
2^{40}	16384	0.5632	–	0.5705

TABLE 1. The expected number $V(n,L)$ of visits to the window $W(n,L)$, the lower bound $h_{n-1}/(10\log(8n/L))$ of Corollary 3.2, and the upper bound of Lemma 2.1. A dash marks the rows with $h_{n-1} < 2\log(8n/L)$, where the corollary does not apply.

Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [9] and working together inside the Brain framework [11], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [10], the Math Brain repository [9] and the Brain framework repository [11].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5 and Claude Fable 5.2 (Anthropic), working in Claude Code, found the statements and the proofs, wrote and ran the program of Section 5, and wrote the paper; GPT-6.1 (OpenAI), working in Codex, checked the proofs independently. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No external data were used. The program of Section 5 and its output are available at `https://github.com/yspennstate/harmonic-descent-occupation-sums`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [9].

Appendix A. Notation and terminology

Symbol or term	Meaning	First appears
h_r	the rth harmonic number $\sum_{j=1}^{r} 1/j$, with $h_0 = 0$	(1.1)

Symbol or term	Meaning	First appears
$(X_t)_{t\geq 0}$	the harmonic descent chain, with transition probabilities (1.1)	Section 1
$p(m, m-d)$	the probability of a jump from m to $m-d$	(1.1)
$\mathbb{P}_n$, $\mathbb{E}_n$	probability and expectation for the chain started at n	Section 1
$a(n,i)$	occupation probability: the probability that the chain started at n visits the state i	Section 1
$a(i)$	the limit of $a(n,i)$ as $n\to\infty$, given by (1.2)	Section 1
weight, f	a non-negative sequence f on $\{2,3,\dots\}$	Section 1
occupation sum	$\sum_{i=2}^{n} a(n,i)f(i)$, for a weight f and a starting point n	Section 1
limit sum	$\sum_{i\geq 2} a(i)f(i)$, finite or infinite	Section 1
$\sim$, o, O, C	asymptotic notation, with C the constant in $x_n = O(y_n)$	Section 1
γ_*	the exponent $1.567\dots$ of [4]	Section 1
K	a constant with $f(i)\leq K/\log i$	Theorem 1.1
φ	a positive sequence tending to infinity	Theorem 1.2
N_j	the starting points of the constructions	Theorem 1.2
$g(i)$	$(i-1)/h_{i-1}$, the expected size of the jump out of the state i	(1.3)
$W(n,L)$	the window $\{n-L,\dots,n-1\}$ of width L below n	Section 1
$V(n,L)$	the expected number of visits of the chain started at n to $W(n,L)$	(1.4)
B	a set of states in $\{2,\dots,n-1\}$	(2.1)
ρ	h_L/h_{n-L-1}	proof of Lemma 2.1
$S_t(i)$	$\sum_{\ell=i}^{i+t} a(\ell,i)$	Lemma 2.2
ε	a positive exponent in Section 1; the splitting fraction in Corollary 2.4(b) and the proof of Theorem 1.1	Section 1
M	the cut-off of the splitting in the proof of Theorem 1.1	proof of Theorem 1.1
θ	a fixed exponent $\theta<1$	Remark 2.6
D_s	the sth jump $X_{s-1}-X_s$ of the chain	Section 3
y, T, q	a bound on the jump sizes, a number of jumps, and $q=h_y/h_{n-1}$	Lemma 3.1
$\mathbf{1}\{\cdot\}$	the indicator of an event	proof of Lemma 3.1
Z	the sum of the jumps of size at most y made from states $\geq n-L$ in the first T steps	proof of Lemma 3.1
E	the event that each of the first T jumps has size at most y	proof of Lemma 3.1
τ	the first time the chain is below $n-L$	proof of Lemma 3.1
λ	$\log(8n/L)$	Corollary 3.2
H	h_{n-1}	proof of Corollary 3.2
x	λ/H	proof of Corollary 3.2
c	the limit of $f(i)/i$	Proposition 1.3
η	a tolerance $\eta>0$	proof of Proposition 1.3

Symbol or term	Meaning	First appears
ψ	$\psi(i) = \inf_{k \geq i} \varphi(k)$	Section 4
L_j, B_j	the widths and the windows $B_j = W(N_j, L_j)$ of the constructions	Section 4
λ_j	$\log(8N_j/L_j)$	Section 4

References

[1] D. Aldous and S. Janson, *The critical beta-splitting random tree II: overview and open problems*, arXiv:2303.02529v3, 2025.

[2] D. Aldous, S. Janson and X. Li, *The harmonic descent chain*, Electron. Commun. Probab. **29** (2024), Paper No. 77, 1–10. https://doi.org/10.1214/24-ECP645, arXiv:2405.05102.

[3] D. Aldous and B. Pittel, *The critical beta-splitting random tree I: heights and related results*, Ann. Appl. Probab. **35** (2025), no. 1, 158–195. https://doi.org/10.1214/24-AAP2112, arXiv:2302.05066.

[4] A. Brandenberger, B. Chin and E. Mossel, *Asymptotics for the harmonic descent chain and applications to critical beta-splitting trees*, arXiv:2505.24821v3, 2026.

[5] Y. R. Chorbadzhiyska, M. Minchev and M. Savov, *Asymptotics for beta-splitting trees via homogeneous fragmentations and meromorphic potential theory*, arXiv:2608.18320v1, 2026.

[6] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. Fernández del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke and T. E. Oliphant, *Array programming with NumPy*, Nature **585** (2020), 357–362. https://doi.org/10.1038/s41586-020-2649-2

[7] A. Iksanov, *The harmonic descent chain and regenerative composition structures*, Electron. Commun. Probab. **30** (2025), Paper No. 11, 1–3. https://doi.org/10.1214/25-ECP661, arXiv:2412.06826.

[8] S. P. Svihla, *Exact asymptotics for the harmonic descent chain and extended convergence bounds for recursive statistics*, arXiv:2611.00347v1, 2026.

[9] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, https://github.com/yspennstate/math-brain-public.

[10] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24, archived on Zenodo, https://doi.org/10.5281/zenodo.24150837.

[11] Y. Shmalo, *Brain Framework*, public software repository, 2026, https://github.com/yspennstate/brain-framework-public.

Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
Email address: yitzchak.shmalo@gmail.com